CRYPTO MEDIUM

NullVector Ransomware

Container Layout

Challenge artifacts: `classified_report.locked` (886 bytes) and `quantum_shor_factors.json`. Inspecting `classified_report.locked`:

- `0x000 - 0x005` (6 B): Magic header `89 52 57 01 01 00`
- `0x006 - 0x105` (256 B): RSA-2048 encrypted session key
- `0x106 - 0x115` (16 B): AES Initialization Vector (IV)
- `0x116 - 0x375` (608 B): AES-256-CBC ciphertext

Layer 1: RSA-2048 Private Key Reconstruction

`quantum_shor_factors.json` supplies public modulus N, public exponent e, and primes p, q. Computing private exponent d: $\phi(N) = (p - 1)(q - 1)$ $d \equiv e^{-1} \pmod{\phi(N)}$
Decrypting the 256-byte wrapped block: $\text{key_block} = \text{wrapped}^d \bmod N$ Taking the trailing 32 bytes extracts the AES-256 key: `078440dd9f8c28d0a8e90f37efd987f736e67023762a2bfbe32cbac9d7afa8d7`

```
[*] Inspecting classified_report.locked (886 bytes):
    Offset 0x000: Magic header [89 52 57 01 01 00]
    Offset 0x006: RSA-2048 encrypted session key (256 bytes)
    Offset 0x106: AES-CBC IV (16 bytes)
    Offset 0x116: Ciphertext payload (608 bytes)

[*] quantum_shor_factors.json loaded: N, e, p, q primes present
[+] RSA private exponent d = e^-1 mod phi(N) recovered
[+] Recovered AES-256 Key: 078440dd9f8c28d0a8e90f37efd987f736e67023762a2bfbe32cbac9d7afa8d7
```

Container Layout and Key Recovery

Layer 2: AES-256-CBC Decryption

Decrypting the 608 ciphertext bytes using the recovered key and IV, then stripping 11 bytes of PKCS#7 padding (`0x0b`), yields a 597-byte intermediate stream.

Layer 3: Affine Byte-Level Obfuscation

Analyzing the 597-byte stream revealed identical 46-byte repeated patterns at offsets 38 and 550, characteristic of standard report dividers (e.g. `==========`).
Testing affine LCG parameters and bitwise rotations over the 46 known bytes: $P[i] = \text{ror}_8(C[i], r) \oplus ((a \cdot i + b) \bmod 256)$

Unique parameter match:

- Rotation: 5 bits right (`r = 5`)
- Divider character: `'='` (`0x3D`)
- Linear mask: $a = 13$, $b = 122 \implies \text{mask}[i] = (13i + 122) \bmod 256$

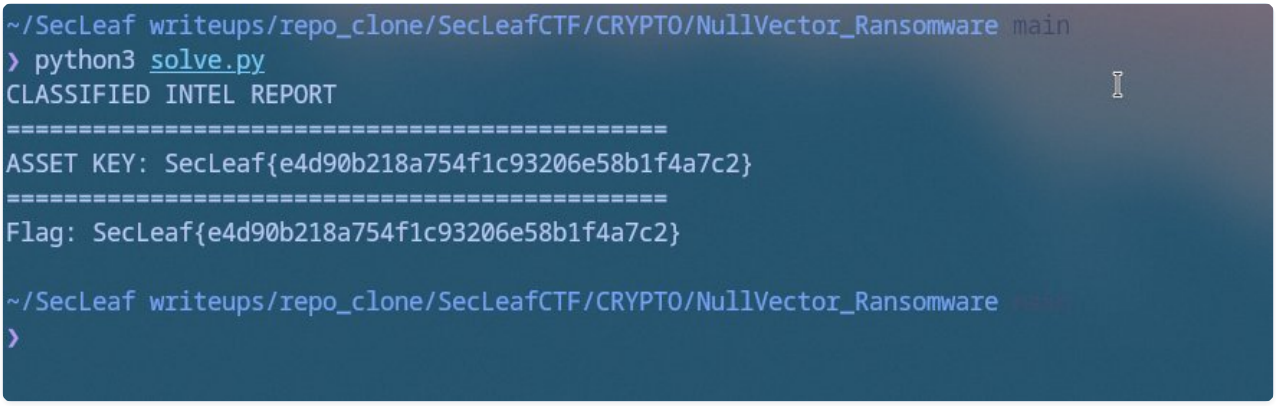
Automated Decryptor

Solution Script: NullVector_Ransomware.py (included in attached solution archive)

Decrypted Report Excerpt & Flag

Executing `solve.py` to decrypt the payload and extract the asset key:

```
$ python3 solve.py
```



Flag Execution Proof

Solution Script: NullVector_Ransomware.py (included in attached solution archive)

Flag:

Solution Script: NullVector_Ransomware.py (included in attached solution archive)

RECOVERED FLAG **SecLeaf{e4d90b218a754f1c93206e58b1f4a7c2}**